LONG-TERM BEHAVIOR

In dynamical systems, eigenvalues determine long-term behavior. For the discrete system $\mathbf{x}_{n+1} = A\mathbf{x}_n$ the deciding number is the absolute value of the eigenvalues.

DISCRETE DYNAMICAL SYSTEM

$$\mathbf{x}_{n+1} = A\mathbf{x}_n$$

split by $|\lambda|$ compared with 1

↑

CASE 1

$|\lambda| > 1$

Grows

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CASE 2

$|\lambda| < 1$

Decays

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CASE 3

$|\lambda| = 1$

Oscillates

CONTINUOUS TIME

For $\mathbf{x}' = A\mathbf{x}$ the solutions involve $e^{\lambda t}$, so the real parts of the eigenvalues decide growth or decay and the imaginary parts decide the oscillation frequency.